

Question number	Answer	Mark
4(a)(i)	The only correct answer is B. <i>A is incorrect because oxygen is not a greenhouse gas</i> <i>C is incorrect because oxygen is not a greenhouse gas</i> <i>D is incorrect because oxygen is not a greenhouse gas</i>	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An answer that includes the following points: • (anthropogenic) caused by the effects of humans (1) • (climate change) changes to (mean) {temperature / rain fall} (1)	IGNORE named activities ACCEPT long-term (mean) change in weather patterns IGNORE weather unqualified / global warming / results of global warming / climate	(2)

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that includes the following points: - species distributed more towards the North (1) - this distribution related to a described change in climate (1) - the effect of this change explained (1)	ACCEPT move away from the equator (northwards) / move towards the (North) pole IGNORE upwards e.g. because the temperatures have become too hot, move to a cooler area, areas have become drier, areas have become wetter, drought e.g. so the enzymes do not work effectively enough (to sustain that species), their prey have migrated (North), they would become dehydrated, plants they feed on die, lack of food	(3)

Question number	Answer	Additional guidance	Mark
4(c)(i)	A description that includes the following points: - temperature (that a beetle was kept at) affects males more than females (1) - the optimum temperature (for keeping both beetles) is 35°C (1)	ACCEPT temperatures higher than 35°C causes a greater decrease in offspring in males than females ACCEPT males at 35°C and females at 38°C / optimum is between 35°C and 38°C	(2)

Question number	Answer	Additional guidance	Mark
4(c)(ii)	An explanation that includes three of the following points: • there are fewer (successful) fertilisations by males / (successful) fertilisations are not (as) affected in females (1) • the sperm are damaged by higher temperatures / egg cells are not (so) affected by higher temperature (1) • higher temperatures decrease sperm {viability / motility} (1) • (higher) temperatures could affect {the acrosome reaction / sperm enzymes / male enzymes} / (higher) temperatures do not (really) affect {egg cell / female} enzymes (1)	ACCEPT offspring for fertilisation ACCEPT higher temperatures make the egg cell easier to penetrate ACCEPT enzymes of females have a higher optimum temperature than the enzymes of {sperm / males} NB If no other marks awarded, credit a correct description of the effect of temperature on enzyme activity	(3)

